
Formatting Instructions For NeurIPS 2025

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Abstract

1

2 1 Introduction

3 In the high stakes world of algorithmic decision making, the outcomes of machine learning models
4 can have significant impacts on individuals and society in areas such as healthcare, criminal justice,
5 lending, hiring. In this contexts, fairness, is not just a desirable property but a critical requirement
6 TODO: [ADD CITATIONS]. In particular, group fairness, which requires predictive behaviour
7 remains equitable across demographically defined groups, has emerged as a key focus of research in
8 this area.

9 Notwithstanding the significant progress in developing group fairness algorithms, most of the existing
10 work assumes a stationary data distribution which, in the real world, rarely holds. Instead, data
11 distributions evolve over time: policy shifts, seasonal effects and changes in societal norms can lead
12 to distribution shifts (often referred to as concept drift) that can undermine predictive performance
13 and fairness guarantees.

14 While online learning algorithms have been created to preserve predictive accuracy, there has been
15 less work on how to preserve fairness under concept drift. This is a critical gap, because fairness
16 violations may remain undetected longer than accuracy degradation, disrupting the trustworthiness of
17 machine learning systems and leading to significant harms.

18 To address this challenge it is important not only to detect distributional changes as they occur but, at
19 the same time, characterize their impact on fairness and react accordingly. In this paper, we study
20 fairness-aware learning under concept drift in online settings and investigate how a novel framework
21 for detecting and responding to concept drift can be used to preserve fairness guarantees over time.

22 2 Experiments

23 Tables 1–5 report per-scenario results on Adult with mean $\pm$ standard deviation across seeds from
24 files/experiments. Aranyani results use 4 available seeds; ARF and RFR use 5 seeds.

Table 1: No drift scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8538 $\pm$ 0.0041	0.0860 $\pm$ 0.0026
ARF	0.7700 $\pm$ 0.0000	0.0336 $\pm$ 0.0000
RFR	0.8050 $\pm$ 0.0079	0.0585 $\pm$ 0.0073

Table 2: Abrupt gender scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8607 ± 0.0199	0.0414 ± 0.0192
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000
RFR	0.8030 ± 0.0045	0.0622 ± 0.0042

Table 3: Gradual gender scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8573 ± 0.0026	0.0129 ± 0.0073
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000
RFR	0.8040 ± 0.0074	0.0690 ± 0.0118

Table 4: Gender-relationship decouple scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8516 ± 0.0052	0.0854 ± 0.0065
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000
RFR	0.7890 ± 0.0096	0.0713 ± 0.0142

Table 5: Occupation gender reversal scenario results on Adult (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	0.8516 ± 0.0056	0.0524 ± 0.0060
ARF	0.7700 ± 0.0000	0.0336 ± 0.0000
RFR	0.7990 ± 0.0124	0.0728 ± 0.0057

Table 6: Folktables dataset results (mean $\pm$ std over seeds)

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani		
ARF	0.735 ± 0.0000	0.0750 ± 0.0000
RFR	0.747 ± 0.0175	0.0257 ± 0.0168

25 **References**

26 References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level
27 heading for the references. Any choice of citation style is acceptable as long as you are consistent. It
28 is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the
29 Reference section does not count towards the page limit.

- 30 [1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In
31 G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp.
32 609–616. Cambridge, MA: MIT Press.
- 33 [2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the*
34 *GENeral NEural Simulation System*. New York: TELOS/Springer-Verlag.
- 35 [3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent
36 synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

37 **A Technical Appendices and Supplementary Material**

38 Technical appendices with additional results, figures, graphs and proofs may be submitted with
39 the paper submission before the full submission deadline (see above), or as a separate PDF in the
40 ZIP file below before the supplementary material deadline. There is no page limit for the technical
41 appendices.

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47 towards the page limit.

48 Please read the checklist guidelines carefully for information on how to answer these questions. For
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- 50 • You should answer [Yes], [No], or [NA].
- 51 • [NA] means either that the question is Not Applicable for that particular paper or the
52 relevant information is Not Available.
- 53 • Please provide a short (1–2 sentence) justification right after your answer (even for NA).

54 **The checklist answers are an integral part of your paper submission.** They are visible to the
55 reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it
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57 with the paper.

58 The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation.
59 While "[Yes]" is generally preferable to "[No]", it is perfectly acceptable to answer "[No]" provided a
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64 write a justification to elaborate. All supporting evidence can appear either in the main paper or the
65 supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification
66 please point to the section(s) where related material for the question can be found.

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- 68 • **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”,**
- 69 • **Keep the checklist subsection headings, questions/answers and guidelines below.**
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72 Question: Do the main claims made in the abstract and introduction accurately reflect the
73 paper’s contributions and scope?

74 Answer: [TODO]

75 Justification: [TODO]

76 Guidelines:

- 77 • The answer NA means that the abstract and introduction do not include the claims
78 made in the paper.
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80 contributions made in the paper and important assumptions and limitations. A No or
81 NA answer to this question will not be perceived well by the reviewers.
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83 much the results can be expected to generalize to other settings.
- 84 • It is fine to include aspirational goals as motivation as long as it is clear that these goals
85 are not attained by the paper.

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87 Question: Does the paper discuss the limitations of the work performed by the authors?

88 Answer: [TODO]

89 Justification: [TODO]

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